

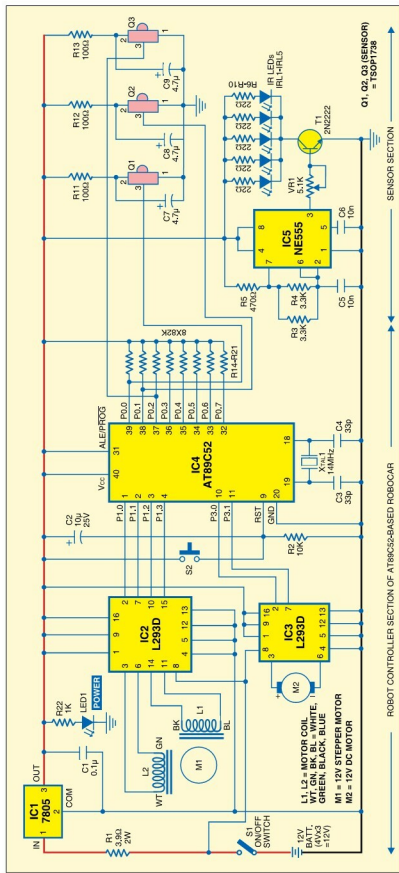


Fig. 2: Schematic diagram of AT89C52-based RoboCar

any obstacle is detected by the IR sensors. It drives the stepper motor (M1) so that the car changes track to avoid a collision with the obstacle.

Port 0 (P0.0 through P0.7) of the microcontroller is used as the input port that is connected to the sensors. Each of these port pins (P0.0 to P0.7) is pulled high through pull-up resistors R14-R21. Note that port pins P0.3 through P0.7 are not used in this application. You can use them to extend the application to make it a line following robot by using similar sensors and making some changes in the code. Port 1 (P1.0 through P1.3) is used for driving the stepper motor through the driver IC L293D (IC2) and port 3 (P3.0 and P3.1) is used for driving the DC motor (M2) for forward or backward motion through another L293D (IC3) driver IC.

Each sensor used here is a 3-pin modular device, where the first and second pins are negative and positive supply terminals respectively and the third pin is the output terminal. A filter circuit comprising a 100-ohm resistor and a 4.7µF capacitor is used for powering each sensor. Normally, the output of the sensor is high. When an IR pulse of frequency 38 kHz falls on the IR sensor, its output goes low. This low output is sensed by Port 0 of the microcontroller and an action is taken as per preprogrammed instructions.

Pin 31 of the microcontroller has to be pulled high so that it can fetch the codes from its internal memory. But if any external memory is used in the circuit, the pin should be pulled low.

Here, R2 and C2 are used for the power-on reset function. As soon as the power is switched on, a high pulse is applied to pin 9 of the microcontroller to reset it. This action initiates the program and the RoboCar starts moving forward.

How a RoboCar works

As soon as the circuit is powered on, the controller IC4 fetches the codes from its internal memory and sets its pin 10 high and pin 11 low. This signal